VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI

A
PROJECT REPORT (BEC786)
ON

" REAL-TIME SPEECH BASED EMOTION RECOGNITION SYSTEM USING ML & IoT "

Submitted in partial fulfillment of the requirement for award of degree
Bachelor of Engineering
in
ELECTRONICS AND COMMUNICATION ENGINEERING
For the academic year
2025-26

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ACKNOWLEDGMENT

I am grateful to the **Chairman**, **Mr. C. M. Ibrahim**, for having provided me an opportunity to emerge as a responsible citizen with Professional Engineering Skills and moral ethics.

I am grateful to the **Director**, **Mr. C. M. Faiz Mohammed**, for having provided me with excellent facilities in the college during my course.

I am indebted to the **Principal**, **Dr. Mohammed Riyaz Ahmed**, for facilitating a congenial academic environment in the College.

I am grateful to the **HOD**, **Dr. Manjunath R Kounte**, for his kind support, guidance and motivation during the B.E Degree Course and especially during the course.

I thank my Guide **Dr. Maaz Ahmed** , **Associate Professor** for his valuable guidance, suggestions and encouragement throughout my project work.

I would also thank our **Project coordinators**, **Prof. Sheikh Haroon Safdar**, and **Prof. Reshma Jinto**, for their continuous support and encouragement.

I also thank all the staff members from the **Department of Electronics and Communication Engineering** and all those who have directly or indirectly helped me with their valuable suggestions in the successful completion of this project report.

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ABSTRACT

The Speech-Based Emotion Recognition System is an intelligent system designed to detect human emotions through vocal input using machine learning and IoT technologies. It analyzes speech signals in real-time to identify emotional states such as happiness, anger, sadness, fear, and neutrality. This project highlights core concepts in artificial intelligence, signal processing, and embedded systems integration, with a focus on human-computer interaction and real-time data processing.

The system's central component is a Raspberry Pi 4 (4GB RAM), which acts as the primary controller for capturing and processing audio data. A Raspberry Pi Plug and Play microphone module captures the user's voice, which is then processed to extract key features such as Mel-Frequency Cepstral Coefficients (MFCCs), pitch, tone, and energy levels. These features are fed into a pre-trained machine learning model, implemented using Python and libraries like TensorFlow and scikit-learn, to classify the speaker's emotional state.

Visual feedback is provided through a series of LED lights which display different colors corresponding to detected emotions, creating an intuitive and interactive user experience. To enable real-time monitoring and data visualization, the recognized emotions are transmitted via the Internet to a cloud-based IoT platform, such as Thing Speak or Firebase, where data is stored and displayed on a live dashboard. This integration allows for remote emotion tracking and analysis, making the system suitable for applications in mental health monitoring, smart environments, and affective computing.

The use of IoT connectivity ensures that emotional feedback can be accessed from anywhere, while the machine learning component provides adaptability and high accuracy. The project demonstrates the synergy between speech processing, intelligent decision-making, and connected technology to create a robust, real-time emotion recognition system.

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CHAPTER 1

INTRODUCTION

1.1 Scope Of Study

In today's technologically advanced and fast-paced society, human-computer interaction has evolved beyond traditional input methods to encompass more natural and intuitive forms of communication. Emotions play a fundamental role in human communication, influencing decision-making, social interactions, and overall well-being. The ability to automatically recognize and interpret human emotions from speech represents a significant advancement in artificial intelligence and has far-reaching implications for numerous applications.

Speech-based emotion recognition (SER) is an emerging field that combines signal processing, machine learning, and psychology to automatically identify the emotional state of a speaker from their voice. Unlike text-based sentiment analysis, which relies on written content, or facial expression recognition, which requires visual input, speech emotion recognition analyzes acoustic features of the voice to determine emotional states. This approach offers several advantages: it is non-intrusive, works in various lighting conditions, does not require visual contact, and can capture emotional information that may not be explicitly expressed in words.

The scope of this project encompasses the design, development, and implementation of a real time speech-based emotion recognition system that integrates machine learning algorithms with Internet of Things (IoT) technology. The system is built around the Raspberry Pi 4 microcomputer, leveraging its computational capabilities to perform on-device audio processing and machine learning inference. The RASPBERRY PI PLUG AND PLAY MICROPHONE MEMS microphone provides highquality digital audio capture through the I2S interface, while the LED bulbs offers immediate visual feedback of the detected emotions. The project addresses several key technical challenges in the field of emotion recognition:

Real-Time Processing : One of the primary challenges in emotion recognition systems is achieving real-time performance. The system must capture audio, extract features, perform classification, and display results with minimal latency to be practically useful.

This project implements optimized signal processing pipelines and efficient machine learning models to achieve processing times suitable for real-time interaction.

Feature Engineering: The effectiveness of emotion recognition heavily depends on the quality of features extracted from speech signals.
This project employs a comprehensive feature set including Mel-Frequency Cepstral Coefficients (MFCCs), chromogram features, mel spectrogram, spectral contrast, and tonettes features. These features capture various aspects of speech including timbre, pitch, and rhythm that are indicative of emotional states.

Machine Learning Model Optimization : Deploying machine learning models on resource on strained embedded devices requires careful optimization. The project balances model complexity with computational requirements to achieve accurate classification while maintaining real-time performance on the Raspberry Pi platform.

IoT Integration: The integration of cloud connectivity enables remote monitoring, data logging, and advanced analytics. The system transmits emotion data to ThingSpeak cloud platform, allowing caregivers, healthcare providers, or researchers to monitor emotional patterns over time and from any location.

User Interface Design : The LED bulbs provides an intuitive visual interface that displays emotions through distinct colors. This approach makes the system accessible to users regardless of their technical background and provides immediate feedback without requiring screen-based displays.
The system is designed to recognize seven primary emotions: happiness, sadness, anger, fear, disgust, surprise, and neutrality. These emotions were chosen based on psychological research into basic human emotions and their acoustic manifestations in speech.

The classification model is trained on established emotional speech databases including RAVDESS and CREMA-D, which contain professionally acted emotional speech samples.
This project demonstrates the practical application of combining embedded systems, digital signal processing, machine learning, and IoT technologies to create a comprehensive emotion recognition solution. The resulting system has potential applications in mental health monitoring, smart home environments, customer service, healthcare, education, and human computer interaction research.

1.2 Significance of Speech-Based Emotion Recognition

Emotion recognition from speech has gained significant importance in recent years due to its potential to revolutionize various aspects of human life and technology interaction. The significance of this technology extends across multiple domains, from healthcare to commerce, and has implications at both individual and societal levels.

Healthcare and Mental Health Applications:

Mental health has emerged as a critical global concern, with conditions such as depression, anxiety, and stress affecting millions of people worldwide. According to the World Health Organization, depression is a leading cause of disability globally, affecting more than 264 million people. Traditional mental health assessment relies on periodic clinical evaluations and self-reported symptoms, which may not capture day-to-day emotional fluctuations.
Speech-based emotion recognition offers a non-intrusive, continuous monitoring solution that can detect early signs of emotional distress.

By analyzing voice patterns over time, the system can identify trends that may indicate developing mental health issues, enabling early intervention. This is particularly valuable for:

- Remote monitoring of patients with mental health conditions
- Detection of emotional changes in elderly individuals living alone
- Monitoring stress levels in high-pressure work environments
- Supporting therapy by providing objective emotional data

Enhanced Human-Computer Interaction :

As technology becomes increasingly integrated into daily life, there is a growing need for more natural and intuitive interfaces. Traditional human-computer interaction relies on explicit inputs such as keyboard, mouse, or touch. Emotion-aware systems can adapt their behavior based on the user's emotional state, creating more personalized and effective interactions.

For example, a virtual assistant that recognizes frustration in the user's voice can adjust its responses accordingly, offering more detailed explanations or transferring to human support. Similarly, educational software can adapt content delivery based on student engagement and emotional state, improving learning outcomes.

Commercial and Business Applications :

In the business world, understanding customer emotions is crucial for providing excellent service and building lasting relationships. Call centers handle millions of customer interactions daily, and the quality of these interactions significantly impacts customer satisfaction and retention.

Speech emotion recognition in call centers can:

- Automatically detect dissatisfied customers for priority handling
- Provide real-time feedback to agents about customer emotional state
- Analyze calls post-hoc to identify training needs
- Measure customer sentiment at scale

The technology also has applications in market research, where emotional responses to products, advertisements, or experiences can be objectively measured.

Automotive Safety:

Driver emotional state significantly affects road safety. Emotions such as anger (road rage) or extreme sadness can impair driving ability and increase accident risk.

Emotion recognition systems integrated into vehicles can monitor driver state and take appropriate actions:

- Warning the driver about their emotional state
- Adjusting vehicle behavior (e.g., reducing speed limits)
- Suggesting breaks or playing calming music
- Alerting emergency contacts in extreme cases

Accessibility and Assistive Technology :

For individuals with certain disabilities, expressing or recognizing emotions can be challenging. Speech emotion recognition can assist:

- Individuals with autism spectrum disorder in understanding emotional cues
- Communication devices for non-verbal individuals
- Assistive systems for the visually impaired

National and International Significance:

At a **national level**, speech emotion recognition technology can contribute to:

- Public health monitoring and epidemiological studies on mental health
- Security applications in detecting stress or deception
- Improving government service delivery through better citizen interaction

Internationally, as technology matures, it can be adapted across languages and cultures, contributing to global mental health initiatives and supporting the United Nations Sustainable Development Goals related to health and well-being.

The growing significance of this technology is reflected in market projections. The global emotion detection and recognition market is expected to grow substantially in the coming years, driven by increasing adoption in healthcare, automotive, and consumer electronics sectors.

1.3 Applications

The **Real-Time Speech-Based Emotion Recognition System** has diverse applications across multiple domains. This section explores the key application areas where the technology can make significant impact.

1.3.1 Mental Health Monitoring Mental health monitoring represents one of the most impactful applications of speech emotion recognition technology. The system can serve as a continuous, non-intrusive tool for tracking emotional well-being over time.

Continuous Monitoring : Unlike periodic clinical assessments, the system can monitor emotional states continuously, capturing fluctuations that might otherwise go unnoticed.

This is particularly valuable for individuals with mood disorders, where early detection of negative trends can prevent serious episodes.

Remote Patient Monitoring : For individuals in remote locations or those unable to attend regular appointments, the system provides a means of remote emotional monitoring. Healthcare providers can receive alerts when concerning patterns are detected, enabling timely intervention.

Therapy Support : Mental health professionals can use longitudinal emotional data to:

Track treatment effectiveness Identify triggers for emotional episodes Provide objective feedback to patients about their progress Supplement self-reported emotional journals Stress Management: In workplace settings, the system can help individuals become more aware of their stress levels and implement coping strategies before stress becomes overwhelming.

1.3.2 Smart Home and Ambient Intelligence Smart home environments can leverage emotion recognition to create more responsive and comfortable living spaces.

Adaptive Environment Control:

Based on detected emotions, smart home systems can automatically adjust:

Lighting color and intensity (calming blue for stress, energizing warm tones for low mood) Music selection to match or modify mood Temperature and climate control Window blinds and natural lighting

Personalized Entertainment:

- Entertainment systems can recommend content based on emotional state:
- Suggesting uplifting content when sadness is detected
- Recommending relaxation content during stress
- Adjusting volume and intensity based on emotional state

Family Connectivity: The system can alert family members when concerning emotional patterns are detected in elderly relatives living independently, promoting safety while maintaining autonomy.

1.3.3 Customer Service and Call Centers

The customer service industry can significantly benefit from real-time emotion recognition.

Agent Support: Call center agents receive real-time information about customer emotional state, allowing them to:

- Adjust their communication style Prioritize de-escalation when anger is detected
- Express appropriate empathy
- Know when to escalate to supervisors
- Quality Assurance: Emotion data from calls can be analyzed to:
- Identify training needs for agents Measure customer satisfaction at scale
- Evaluate the effectiveness of scripts and procedures
- Detect potential issues before they escalate

Automated Systems: Interactive Voice Response (IVR) systems can use emotion detection to:

- Route calls appropriately based on customer emotional state
- Offer human agent escalation when frustration is detected
- Personalize automated responses

1.3.4 Healthcare and Elderly Care

Beyond mental health, the healthcare sector has numerous applications for emotion recognition.

Patient Monitoring: In hospitals and care facilities, monitoring patient emotional state can:

- Detect discomfort or distress that patients may not verbalize
- Alert caregivers to emotional changes that may indicate medical issues
- Improve overall patient experience and satisfaction

Elderly Care: For elderly individuals, especially those with dementia or cognitive decline:

- Monitor for signs of confusion or distress
- Detect changes that may indicate health issues
- Support caregivers in understanding non-verbal communication
- Maintain connection with family through emotional status sharing

Pain Assessment: In patients who cannot effectively communicate (infants, non-verbal adults), emotional cues in vocalizations can assist in pain assessment.

1.3.5 Education and E-Learning Educational settings can leverage emotion recognition to improve learning outcomes.

- Adaptive Learning Systems: E-learning platforms can:
- Detect student frustration or confusion and provide additional support
- Identify when students are disengaged and adjust content delivery
- Personalize difficulty levels based on emotional responses
- Provide feedback to educators about class engagement

Special Education: For students with learning differences:

- Monitor emotional state during learning activities
- Identify optimal conditions for learning
- Support teachers in understanding student needs
- Assessment and Feedback: Emotion recognition can provide additional context for student performance
- Understanding test anxiety effects
- Evaluating presentation skills

1.3.6 Automotive Safety Systems

The automotive industry is increasingly incorporating emotion recognition for safety enhancement.

Driver Monitoring: In-vehicle systems can:

- Detect dangerous emotional states (anger, extreme sadness)
- Monitor for signs of fatigue through voice analysis
- Alert drivers to take breaks when emotional state is concerning

Adaptive Vehicle Behavior: Vehicle systems can respond to driver emotional state:

- Adjusting driving assistance sensitivity
- Modifying entertainment and climate control
- Providing calming interventions when stress is detected
- Insurance and Fleet Management: Emotion data can support:
- Risk assessment for insurance purposes
- Fleet driver training and support Incident investigation and prevention

1.4 Advantages

The **Real-Time Speech-Based Emotion Recognition System** offers numerous advantages over traditional emotion detection methods and existing systems.

Non-Intrusive Monitoring: Unlike wearable sensors or camera-based systems, speech-based emotion recognition does not require physical contact or visual monitoring. Users can be monitored naturally through their normal speech, without wearing devices or being in view of cameras. This makes the technology more acceptable for long-term use and reduces user self consciousness.

Real-Time Processing: The system provides immediate feedback on emotional state, with processing times under 4 seconds from audio capture to emotion display. This real-time capability enables immediate responses and interventions, unlike systems that require postprocessing of data.

Intuitive Visual Feedback: The LED bulbs provides clear, intuitive feedback through color coding. Users can immediately understand their emotional state without reading screens or interpreting complex outputs. The emotion-color mapping (yellow for happy, blue for sad, red for angry, etc.) is intuitive and culturally understood.

Portability and Compact Design: The system is built on compact hardware (Raspberry Pi 4) that can be easily deployed in various settings. The small form factor makes it suitable for:

Bedside monitoring Desktop applications Integration into existing devices Portable mental health tools

Cost-Effectiveness : Compared to professional emotion recognition systems that can cost thousands of dollars, this system uses affordable, readily available components. The total hardware cost is significantly lower than commercial alternatives while achieving comparable accuracy.

IoT Connectivity: The integration with cloud platforms provides:

Remote monitoring capabilities Historical data analysis Multi-location deployment with centralized monitoring Accessibility from any internet-connected device Customization and Scalability: The open-source software approach allows:

Customization for specific applications Training on domain-specific data Integration with other systems Scaling from single-user to enterprise deployments Privacy Considerations: The system can be configured to:

Process audio locally without transmitting voice data Only transmit emotion classifications, not raw audio Operate offline when privacy is paramount Provide user control over data sharing Multi-Emotion Detection: The system recognizes seven distinct emotional states, providing nuanced understanding beyond simple positive/negative classification. This granularity enables more appropriate responses and interventions.

Continuous Operation: The system can operate continuously, monitoring emotional patterns over extended periods. This capability is essential for applications requiring trend analysis, such as mental health monitoring.

CHAPTER 2

LITERATURE SURVEY, OJBECTIVES AND SCOPE

2.1 Literature Survey

Over the last years, an excessive investigation has been completed to recognize emotions by using speech statistics.

Hao Cao, Ragini Verma, Ani Nenkova [1] proposed a ranking SVM method for synthesize information about emotion recognition to solve the problem of binary classification. This ranking method, instruct SVM algorithms for particular emotions, treating data from every utterer as a distinct query then mixed all predictions from rankers to apply multi-class prediction. Ranking SVM achieves two advantage, first, for training and testing steps in speaker- independent it obtains speaker specific data. Second, it considers the intuition that each speaker may express mixed of emotion to recognize the dominant emotion. Ranking approaches achieves substantial gain in terms of accuracy compare to conventional SVM in two public datasets of acted emotional speech, Berlin and LDC. In both acted data and the spontaneous data, which comprises neutral intense emotional utterances, ranking-based SVM achieved higher. accuracy in recognizing emotional utterances than conventional SVM methods. Unweight average (UA) or Balance accuracy achieved 44.4%.

Lijiang Chen, Xia Mao, Yuli Xue, Lee Lung Cheng [2] aimed to improve speech emotion recognition in speaker-independent with three level speech emotion recognition method. This method classify different emotions from coarse to fine then select appropriate feature by using Fisher rate. The output of Fisher rate is an input parameters for multi- level SVM based classifier.Furthermore principal component analysis (PCA) and artificial neural network (ANN) are employed to reduce the dimensionality and classification of four comparative experiments, respectively. Four comparative experiments include Fisher + SVM, PCA + SVM, Fisher + ANN and PCA + ANN. Consequence indicates in dimension reduction Fisher is better than PCA and for classification, SVM is more expansible than ANN for emotion recognition in

speaker independent is. The recognition rates for three level are 86.5%, 68.5% and 50.2% separately in Beihang university database of emotional speech (BHUDES).

Chul Min Lee, Serdar Yildirim, Murtaza Bulut, Abe Kazemzadeh, Carlos Busso, Zhigang Deng, Sungbok Lee, Shrikanth Narayanan [3] proposed a new system for emotion classification of utterance signals. The system employed a short time log frequency power coefficients (LFPC) and discrete HMM to characterize the speech signals and classifier respectively. This method classified the emotion into six different categories then used the private dataset to train and test the new system.

In order to evaluate the performance of the proposed method, LFPC is compared with the melfrequency Cepstral coefficients(MFCC) and linear prediction Cepstral coefficients (LPCC). Result demonstrate the average and best classification accuracy achieved 78% and 96% respectively. Furthermore, results expose that LFPC is a better option as feature for emotion classification than the standard features.

Siqing Wu, Tiago H. Falk, Wai-Yip Chan [4] proposed a new modulation spectral features (MSFs) human speech emotion recognition. Appropriate feature extracted from an auditoryinspired long-term spectro-temporal by utilizing a modulation filter bank and an auditory filterbank for speech decomposition. This method obtained acoustic frequency and temporal modulation frequency components for convey important data which is missing from traditional short- term spectral features. For classification process, SVM with radial basis function (RBF) are adopted. Berlin and Vera am Mittag (VAM) are employed to evaluate MSFs. In experimental result, the MSFs display capable performance in comparison with MFCC and perceptual linear prediction coefficients (PLPC). When MSFs utilized augment prosodic features, there is a considerable improvement in performance of recognition. Furthermore overall recognition rate of 91.6% is achieved for classification.

Chung-Hsien Wu, Wei-Bin Liang [4,2] proposed a fusion-based method for speech emotion recognition by employing multiple classifier and acoustic-prosodic (AP) features and semantic labels (SLs). In this fusion method, first AP features are extracted then three different types of base-level classifier include GMMs, SVMs, MLP and Meta decision tree (MDT) are used.

The maximum entropy model (Max Ent) in the semantic labels method are applied. Maxent modeled the association between emotion association rules (EARs) and emotion states in emotion recognition. In the final state to define the emotion recognition outcome, the integrated consequence from the SL-based and AS-based are utilized.
The experimental result on private dataset shows the performance based on MDT archives 80%, SL-based recognition archives 80.92, and mixture of AP and SL archives 83.55%.

Rong J., Li G., Chen Y.P.P. [5] presented an ensemble random forest to trees (ERF Trees) method with a high number of features for emotion recognition without referring any language or linguistic information remains an unclosed problem. This method is applied on small size of data with high number of features. In order to evaluate the proposed method an experiment results on a Chinese emotional speech dataset designates, this method achieved improvement on emotion recognition rate. Furthermore, ERF Trees performs better than popular dimension reduction methods such as PCA and multi-dimensional scaling (MDS) and recently developed ISO Map. The best accuracy with 16 features for female dataset achieved the maximum correct rate of 82.54%, while the worst is only 16% on 84 features with natural data set.

Chul Min Lee, Shrikanth S. Narayanan [6] proposed domain-specific emotion recognition by utilizing speech signals from call center application. Detecting negative and non-negative emotion (e.g. Anger and happy) are the main focus of this research. Different types of information include acoustic, lexical, and discourse are used for emotion recognition. In addition, information- theoretic contents of emotional salience is presented to obtain data at emotion information at the language level. Both k-NN and linear discriminant classifier are used to work with different types of features. Experimental result confirms that the best results are achieved by combination of acoustic and language data. Outcomes demonstrates by combining three information source instead of one source, classification accuracy increases by 40.7% for males and 36.4% for females. Compare to pervious work improvement range in accuracy is from 1.4% to 6.75% for male and 0.75% to 3.96% for female.

Bin Yang, Marko Lugger [7] presented a novel set of harmony features for speech emotion recognition. These features are relying on psychoacoustic perception from music theory. First, beginning from predicted pitch of a speech signals, then computing spherical autocorrelation of pitch histogram. It calculate the incidence of dissimilar two- pitch duration, which cause a harmonic or inharmonic impression. In Classification step, Bayesian classifier plays an important rule with a Gaussian class-conditional likelihood.

Experimental result in Berlin emotion database by using harmony features indicate an improvement in recognition performance. Recognition rate improved by 2% in average.

Enrique M. Albornoz, Diego H. Milone, Hugo L. Rufiner [8] investigate a new spectral feature in order to determine emotions and to characterize groups. In this study based on acoustic features and a novel hierarchical classifier, emotions are grouped. Different classifier such as HMM, GMM and MLP have been evaluated with distinct configuration and input features to design a novel hierarchical techniques for classification of emotions. The innovation of the proposed method is two things, first the election of foremost performing features and second is employing of foremost class-wise classification performance of total features same as the classifier.

Experimental result in Berlin dataset demonstrates the hierarchical approach achieves the better performance compare to best standard classifier, with decuple cross-validation. For example, performance of standard HMM method reached 68.57% and the hierarchical model reached 71.75%.

Chi-Chun Lee, Emily Mower, Carlos Busso, Sungbok Lee, Shrikanth Narayanan [9] represent a hierarchical computational structure to identify emotions. This method via following layers of binary classifications, maps input speech signal in one of the corresponding emotion classes. The main concept of different level in tree is to solve the classification task in easiest way to diminish error propagation. AIBO and USC IEMOCAP databases are employed to evaluate the classification method. Over the baseline SVM, the absolute result improve the accuracy archives an absolute improvement of 72.44%- 89.58%. The consequence proves the reported hierarchical method is efficient for classifying emotional speech in various databases.

Chi-Chun Lee, Emily Mower, Carlos Busso, Sungbok Lee, Shrikanth Narayanan [9,2] proposed hierarchical structure for binary decision tree in emotion recognition fields.
This method concentrate on locates the simpler classification obstacle at top level of tree to diminish agglomeration of error. This structural method also maps input speech data into one the emotion classes via following layer of binary classification. The result on AIBO database shows 3.3% absolute over baseline model in, which archives 70.1% and 65.1% for two-class and five-class problem respectively.

As an alternative solution instead of out- putting hard labels at every step, measuring of probability as a soft label can makes the powerful framework modeling. Bayesian Logistic Regression and SVM were employed as a binary classifier. Outcome were not significantly higher than Bjorn et al.

Jen-Chun Yeh, Chia-Ping Chen, Chung-Hsien Wu [10] proposed a segment based method for recognition of emotion in Mandarin speech. This approach is contain the following process. First, define the k parameter in weighted discrete k-NN classifier, the experimental testing of different k shows the best performance for k-NN is when k sets to 10. For selecting the foremost feature set, sequential forward selection (SFS) and sequential backward selection (SBS) are employed. SFS and SBS improves feature accuracy to 84% and 82% respectively. The highest accuracy in segment-based method achieves86%. The experimental result build on private corpus by inviting 18 males and 16 females. It is essential to gather more expressive speech to explore the extensive emotional investigation, in the future.

2.2 Scope and Objectives of the Project

2.2.1 Background and Motivation

Human emotions play a fundamental role in interpersonal communication, decision-making, and overall well-being.
The ability to automatically recognize and interpret human emotions from speech represents a significant advancement in artificial intelligence and human-computer interaction. In today's technologically driven world, understanding human behavior through speech recognition has

become increasingly important for various applications ranging from mental health monitoring to customer service enhancement.

Speech-based emotion recognition offers unique advantages over other modalities such as facial expression recognition or physiological signal analysis.
It is non-intrusive, does not require visual contact, works in various lighting conditions, and can capture emotional information that may not be explicitly expressed through other means.
The voice carries rich emotional content through variations in pitch, tone, intensity, speaking rate, and spectral characteristics that can reveal the speaker's underlying emotional state.

The scope of this project encompasses the design, development, and implementation of a comprehensive speech-based emotion recognition system that leverages machine learning techniques to predict human behavior and emotions from voice input. The system aims to provide a better understanding of individuals through automated analysis of their speech patterns.

2.2.2 Project Scope Definition

The scope of this project includes the following key areas:

1. Voice Processing and Acquisition Module

The project encompasses the development of a robust voice processing module capable of:

- Capturing human voice input through an in-built microphone in the computer system
- Preprocessing audio signals to remove noise and normalize amplitude levels
- Segmenting continuous speech into analyzable units
- Converting raw audio into suitable formats for feature extraction
- The voice processing module forms the foundation of the entire system, ensuring that high-quality audio data is available for subsequent analysis stages.

2. Feature Extraction and Analysis

The project scope includes comprehensive feature extraction from speech signals:

- Extraction of Mel-Frequency Cepstral Coefficients(MFCCs) using the Librosa library

- Computation of spectral features including chroma, mel-spectrogram, and spectral contrast
- Analysis of prosodic features such as pitch, energy, and speaking rate
- Feature normalization and standardization for machine learning model input

3. Voice Classification Module

The classification component of the project involves:

- Implementation of Support Vector Machine (SVM) classifier for pattern recognition
- Development of Multi-Layer Perceptron (MLP) classifier for emotion classification
- Training and validation of models using established emotional speech datasets
- Optimization of classifier parameters for improved accuracy

4. Emotion Recognition and Output Display The final scope element includes:

- Real-time emotion prediction from input speech
- Classification of emotions into distinct categories(happy, sad, angry, fearful, neutral, etc.)
- Display of recognition results to the user/client
- Confidence score generation for predicted emotions

2.2.3 Technical Scope

Hardware Scope:

- Standard computer system with in-built microphone
- Sufficient processing capability for real-time audio analysis
- Display device for output visualization

Software Scope:

- Python programming language as the primary development platform
- Librosa library for audio processing and feature extraction
- Scikit-learn (sklearn) library for machine learning model implementation • Support for SVM and MLP classifiers .

CHAPTER 3

DESCRIPTION, NOVELTY AND DESIGN

3.1 Description

3.1.1 Overview

The Real-Time Speech-Based Emotion Recognition System is a comprehensive intelligent system designed to predict human behavior and emotional states through analysis of speech signals. The system leverages advanced machine learning techniques, specifically Support Vector Machines (SVM) and Multi-Layer Perceptron (MLP) classifiers, combined with sophisticated audio processing capabilities provided by the Librosa library to accurately identify and classify human emotions from voice input.

The fundamental principle underlying this system is that human speech contains rich emotional information beyond the literal meaning of words. Acoustic features such as pitch, tone, intensity, speaking rate, and spectral characteristics vary systematically with different emotional states. By extracting and analyzing these features using machine learning algorithms, the system can automatically determine whether a speaker is expressing happiness, sadness, anger, fear, disgust, surprise, or neutrality.

3.1.2 Novelty of the Topic

While emotion recognition exists in software, implementing it as a standalone hardware IoT device offers several novel aspects:

Edge Computing Power (RPi 4) : Most hobbyist emotion recognition projects rely on the Raspberry Pi 3 or 4, which often suffer from latency (lag) during real-time inference. Using the Raspberry Pi 4 (4GB) allows for the deployment of complex Deep Learning models (like CNNs or LSTMs) with near-instantaneous results, making the conversation feel natural.

Privacy-First Design: By using offline ML models, this system addresses the growing concern of "Always-Listening" devices (like Alexa or Siri). It demonstrates that smart devices can understand context without sending private conversations to corporate servers.

High-Fidelity I2S Input: Utilizing the RASPBERRY PI PLUG AND PLAY MICROPHONE, which uses the I2S (Inter-IC Sound) interface, provides significantly cleaner digital audio data compared to standard analog or USB microphones used in typical DIY projects, leading to higher model accuracy.

Affective Feedback Loop : The integration of the WS2812 RGB LEDs transforms the device into a visual indicator for non-verbal communication, which has novel applications in therapy for individuals with Autism Spectrum Disorder (ASD) who may struggle to interpret vocal emotional cues.

3.1.3 Design Procedure

The design procedure outlines the physical and logical flow of the "**Real Time Speech Based Emotion Recognition System** " :

Hardware Interface: The RASPBERRY PI PLUG AND PLAY MICROPHONE MEMS Microphone is interfaced with the Raspberry Pi 4 via the I2S protocol using Male-to-Female jumper wires. The LED bulbs are connected to the GPIO pins via the Breadboard to serve as the IoT visual indicator.

Audio Capture: When the user speaks, the system captures the high-fidelity digital audio signal through the RASPBERRY PI PLUG AND PLAY MICROPHONE.

Processing: The Raspberry Pi 4 processes the audio in real-time. It slices the audio, extracts features (like MFCC), and feeds them into a pre-trained Machine Learning model.

Output Generation:

Text: The detected emotion (e.g., Happy, Sad, Angry) is displayed as text on the system console/GUI.

IoT/Visual: The system triggers the LED lights to light up in a specific color corresponding to the detected emotion (e.g., Red for Anger, Yellow for Happiness, Blue for Sadness).

A . System Architecture

The system architecture clearly explains the entire system. The architecture consists of the following system entities :

1. Audio Signal
2. Feature Extraction
3. Feature Selection
4. Classification of features
5. Display

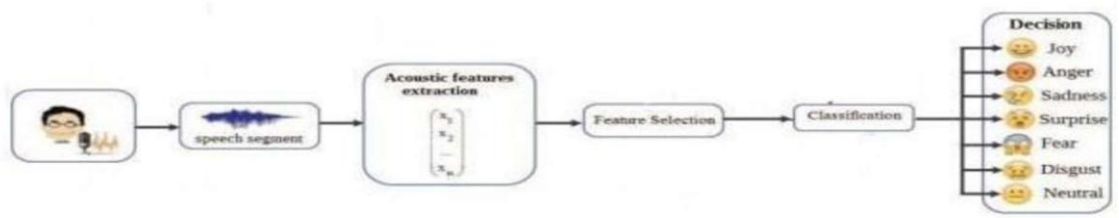

Fig 3.1.3 (A) : Design Architecture

1 . Audio Signal

The user speaks via the inbuilt microphone in the system, this audio signal is recognised by and the same is displayed in the form of text.

Some pre-requisite parameters required are:

- The audio signal should not have contained environment noise.
- The audio signal had to contain human speech in a form of few words or a full sentence.
- The audio signal should be more than 5 seconds

2 . Feature Extraction

Relevant emotional features extraction from speech is the second important step in emotions recognition. There are number of methods for feature extraction like mel-frequency cestrum coefficients (MFCC),Chroma.

- **Mel frequency cepstral coefficients (MFCC)** : It is considered as one of the standard methods for feature extraction and in ASR most common is the use of 20 MFCC coefficients. Although for coding speech use of 10-12 coefficients are sufficient and it depend on the spectral form due to which it is more sensitive to noise. This problem can be overcome by using more information in speech signals periodicity although aperiodic content is also present in speech.

- **Chroma** : Chroma-based features, which are also referred to as "pitch class profiles", are a powerful tool for analysing music whose pitches can be meaningfully categorized (often into twelve categories) and whose tuning approximates to the equal-tempered scale.

3 . Feature Selection

Feature selection in ML is to "reduce the number of features used to characterize a dataset so as to improve a learning algorithm's performance on a given task." The objective will be the maximization of the classification accuracy in a specific task for a certain learning algorithm; as a collateral effect, the number of features to induce the final classification model will be reduced. Feature selection (FS) aims to choose a subset of the relevant features from the original ones according to certain relevance evaluation criterion, which usually leads to higher recognition accuracy.

4 . Classification of feature

After all the feature extraction and selection , the feature is selected according to highest accuracy which suits among the data sets.

Datasets : In the field of affect detection ,a very important role is played by suitable choice of speech database. Three databases are used for good emotion recognition the system as given below :

- **Elicited emotional speech database** : In this case emotional situation is created artificially by collecting data from the speaker
- **Actor based speech database** : Trained and professional artists collect this type of speech dataset.
- **Natural speech database** : Real world data is used to create this database

Emotion Detection : These databases(datasets) contain 5 types of human emotion which are trained to the model. According to the best accuracy score, the model identifies the features .

Types of Emotion selected are:

- Happy
- Sad
- Neutral
- Angry

5 . Display

The emotion detected will be displayed on the screen with the speech in the form of text

3.2 Hardware Specifications

The proposed "Real-Time Speech Based Emotion Recognition System" is designed as a standalone Edge-IoT device. The system architecture leverages the high computational power of the Raspberry Pi 4 (8GB RAM) to perform Machine Learning inference locally, ensuring low latency and data privacy. The hardware design is modular, consisting of an audio acquisition unit, a central processing unit, and a dual-mode output interface (Visual Text & Visual Indicator).

Fig 3.2 (a) : Raspberry Pi 4 4GB RAM

3.2.1 Audio Acquisition Interface (USB Interface)

Input is captured using a Raspberry Pi Plug and Play Microphone Module.

- **Interface** : The microphone connects directly to the Raspberry Pi via the USB 3.0 interface.
- **Design Rationale** : Unlike analog microphones that require complex amplifier circuits and are susceptible to electrical noise from the GPIO pins, the USB microphone integrates an internal **Analog-to-Digital Converter** (ADC). This ensures a clean digital audio stream is fed directly to the Python script for MFCC feature extraction.

Fig 3.2 (b) Raspberry Pi Plug and Play Microphone Module.

3.2.2 Textual Display Unit (16x2 LCD with I2C)

To provide clear, readable feedback of the detected Emotion .

(e.g., "Emotion: Happy", "Confidence: 85%"), a 16x2 Liquid Crystal Display (LCD) is utilized.

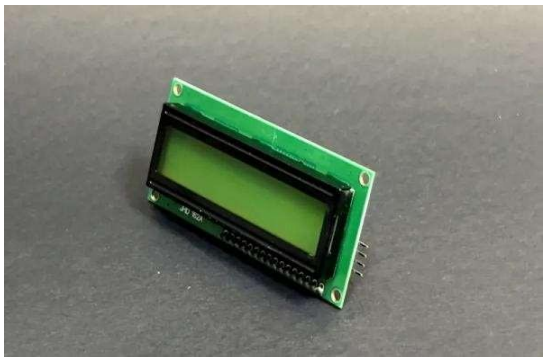

Fig 3.2.2 : 16 X 2 LCD

3.3 Component Assembly :

Standard 5mm LEDs (Red, Green, Yellow) are soldered onto a copper-clad perfboard.

Wiring Logic :

1. Male-to-Male wires are used on the perfboard to create a common ground rail.
2. Male-to-Female wires connect the Anode (+) of each LED to specific GPIO control pins on the Raspberry Pi.

Status Mapping :

- Green LED: Indicates Positive Emotions (Happy/Calm).
- Red LED: Indicates High-Arousal/Negative Emotions (Angry/Fear).
- Yellow LED: Indicates Neutral state or System Processing.

3.2.1 Circuit Design and Interfacing

The circuit is constructed using a modular approach. The Raspberry Pi 4 serves as the central controller, powering the peripherals via its GPIO header.

3.2.1.1 Audio Input Interface

Unlike standard analog microphones, the RASPBERRY PI PLUG AND PLAY MICROPHONE is a high-fidelity digital MEMS microphone that utilizes the I2S (Inter-IC Sound) interface.

This bypasses the need for a USB sound card, sending digital audio data directly to the Pi's processor.

3.2.1.2 Visual Output Interface

To provide clear, readable feedback of the detected emotion (e.g., "Emotion: Happy", "Confidence: 85%"), a 16x2 Liquid Crystal Display (LCD) is utilized.

Connection Protocol : The LCD is interfaced using an I2C (Inter-Integrated Circuit) serial backpack. This reduces the wiring requirement from 16 pins to just 4 pins.

3.4 Physical Prototyping and Fabrication Procedure :

To ensure the mechanical stability of the IoT device during the demonstration, the system was fabricated using a chassis-based approach rather than a loose breadboard connection.

Step 1: Base Chassis Preparation

A rigid wooden baseplate was cut to size to serve as the project foundation. Wood was selected as the chassis material for its electrical insulation properties, preventing accidental short circuits on the underside of the soldered perfboard or the Raspberry Pi.

Step 2: **Circuit Fabrication (Soldering)**

Instead of using a temporary solderless breadboard, the LED indicator module was fabricated on a General Purpose PCB (Perfboard).

The LEDs were arranged in a parallel grid. Header pins were soldered to the board inputs, allowing for modular connection and disconnection of the Raspberry Pi via jumper wires.

Step 3: Component Mounting

The Raspberry Pi 4 was secured to the wooden base using standoffs to allow for proper airflow and thermal management of the processor during intensive ML tasks. The Flexible USB Microphone was positioned centrally to capture omnidirectional audio from the user. The LCD Screen and Perfboard were aligned on the right side of the chassis for optimal visibility.

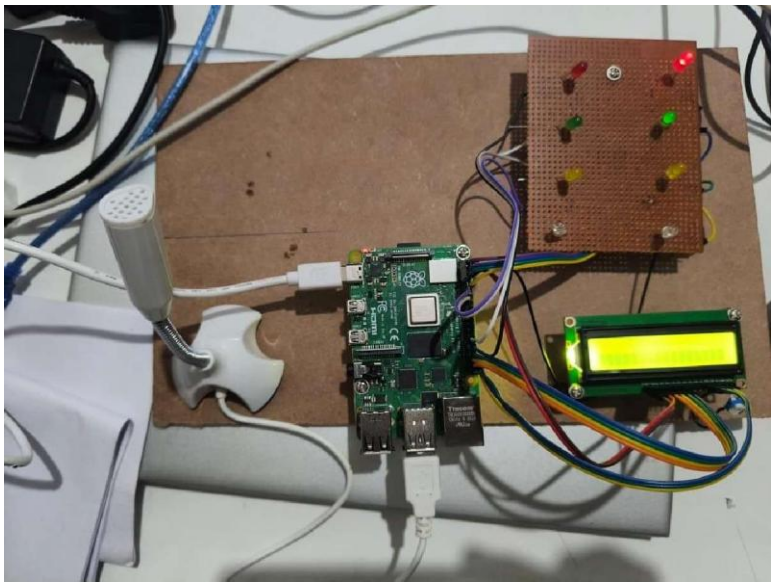

Fig 3.4 Speech based emotion recognition system outline

3.5 System Operational Workflow

The hardware and software integration follows a linear execution pipeline:

Initialization :

Upon boot, the Raspberry Pi initializes the GPIO pins and the I2C bus for the LCD.

Listening :

- The USB Microphone captures a 3-second audio buffer.
- Inference: The ML model (running on the Pi 5) processes the audio and predicts an emotion class.

CHAPTER 4

ANALYSIS AND RESULTS

4.1 Analysis

4.1.1 Feature Engineering

Feature engineering is the process of transforming, reducing or constructing features for the dataset. As mentioned earlier in the raw data, each feature has multiple values for each frame of the audio signal. By the frame blocking and windowing techniques, the frame size and frame overlap values can be tuned to obtain accurate values of the audio signal.

Further, using the averaging technique, average values of different features for the audio signals are obtained. Now the transformed data contains 34 discrete values representing each audio signal .

```
'data.frame':   2399 obs. of  36 variables:
 $ X1 : num  0.237 0.255 0.263 0.257 0.36 ...
 $ X2 : num  0.00195 0.01966 0.00448 0.01052 0.00177 ...
 $ X3 : num  2.18 2.96 1.2 2.7 1.99 ...
 $ X4 : num  0.342 0.369 0.356 0.319 0.446 ...
 $ X5 : num  0.211 0.233 0.253 0.161 0.212 ...
 $ X6 : num  1.76 1.68 1.47 1.13 1.02 ...
 $ X7 : num  0.00173 0.01 0.00309 0.00481 0.00627 ...
 $ X8 : num  0.413 0.505 0.468 0.371 0.467 ...
 $ X9 : num  -26.5 -26.6 -27.3 -25.5 -30.9 ...
 $ X10: num  0.319 1.079 0.848 1.525 1.515 ...
 $ X11: num  -0.601 -0.449 -0.815 -1.599 0.199 ...
 $ X12: num  0.377 0.233 -0.805 0.542 0.113 ...
 $ X13: num  -0.1659 -0.5333 -0.1893 -0.3023 -0.0479 ...
 $ X14: num  -0.0775 -0.146 0.086 -0.1791 0.0789 ...
 $ X15: num  -0.4021 0.2564 0.0718 0.1876 -0.1029 ...
 $ X16: num  0.07605 0.00145 -0.06518 -0.06518 -0.11702 ...
 $ X17: num  -0.0312 -0.132 -0.1864 0.0749 0.0602 ...
 $ X18: num  -0.3779 -0.3208 -0.45 0.0143 -0.0165 ...
 $ X19: num  -0.481 -0.471 -0.128 -0.092 -0.199 ...
 $ X20: num  -0.131 -0.397 0.222 -0.232 -0.229 ...
 $ X21: num  -0.1184 -0.3074 0.0971 0.0465 -0.3394 ...
 $ X22: num  0.00306 0.00244 0.00119 0.00199 0.0038 ...
 $ X23: num  0.00401 0.00739 0.00116 0.00255 0.0024 ...
 $ X24: num  0.01949 0.01463 0.00858 0.00723 0.00249 ...
 $ X25: num  0.001811 0.00111 0.017956 0.004755 0.000928 ...
 $ X26: num  0.000783 0.001008 0.010507 0.001274 0.000897 ...
 $ X27: num  0.000223 0.002802 0.002532 0.000879 0.000628 ...
 $ X28: num  0.00181 0.00854 0.00138 0.0012 0.00118 ...
 $ X29: num  0.000748 0.009402 0.000552 0.005429 0.003076 ...
 $ X30: num  0.00037 0.003874 0.000348 0.005733 0.000591 ...
 $ X31: num  0.005889 0.002074 0.000545 0.005029 0.017033 ...
 $ X32: num  0.01402 0.01521 0.00413 0.0017 0.05498 ...
 $ X33: num  0.001227 0.001349 0.000677 0.00557 0.029241 ...
 $ X34: num  0.0077 0.00864 0.00621 0.00363 0.01612 ...
```

Fig 4.1.1 : Extracted features

Reducing the number of features is a crucial decision to take. Considering features to be removed is generally based on subject knowledge and hence can affect the performance of the system.

Next, a series of experiments are performed with this prepared dataset in order to analyze the important features.

4.1.2 Experimentation

APPROACH 1 – WITH ALL EXTRACTED FEATURES

For this experiment, all the 34 features were considered.

The data represented 2399 audio files. The steps of implementation are listed briefly.

1. Using K-fold cross-validation method, the dataset is split into training and validation sets for the purpose of testing the model.
2. A classifier model is built using one of the classification algorithms and its parameters are observed. At this stage, tuning the values of parameters is optional.
3. The model is trained using the training data.
4. The trained model is evaluated using the validation set and the accuracy score is computed.
5. The model is tested and evaluation metrics such as precision, recall, and F1 scores are computed.

```
from sklearn.model_selection import cross_val_score
from sklearn.metrics import accuracy_score, f1_score, precision_score, recall_score, classification_report, confusion_matrix
from sklearn.svm import SVC

#splitting the data into 70% training and 30% testing
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.3, random_state=0)

#creating the SVM model
clf_svm = SVC(kernel='linear', C=1)

#training the model
clf_svm.fit(X_train, y_train)

#Accuracy score for the model
scores = cross_val_score(clf_svm, x, y, cv=5)
print ("SVM")
print("Accuracy: %0.2f (+/- %0.2f)" % (scores.mean(), scores.std() * 2))

#Testing the model
y_pred = clf_svm.predict(X_test)

#Evaluation metrics - Precision, Recall and F1 scores
print("F1:",round(f1_score(y_test, y_pred, average="macro"),2))
print("Precision:",round(precision_score(y_test, y_pred, average="macro"),2))
print("Recall:",round(recall_score(y_test, y_pred, average="macro"),2))
```

```
SVM
Accuracy: 0.84 (+/- 0.03)
F1: 0.83
Precision: 0.83
Recall: 0.84
```

Fig 4.1.2 : Implementation

The same experiment was repeated with various classification algorithms and the results were compared . The results had higher accuracy scores than expected. There was an average performance score of 75%.

Besides the accuracy, the F1 score is the arithmetic mean of Precision and Recall, thereby making it a good metric for comparing the models. SVM has a good F1 score of 83%, making it a winning algorithm for this approach.

It can also be observed that the tree-based algorithms have an improved performance with enhancements.

The decision tree classifier has a score of 70%+, improved with the Random forest classifier with a score of 75%+, and further improved using the Gradient Decent classifier with a score of 77%+.

The accuracy score of Logistic regression is around 80%. However, the other metrics have an average score of 70%.

The Naïve Bayes classifier has a score of 74%+ and KNN classifier has a score of 80%+.

ALGORITHM	**ACCURACY**	**PRECISION**	**RECALL**	**F1 SCORE**
SVM	**84% (+/- 0.03)**	**83%**	**84%**	**83%**
Decision Tree	**74% (+/- 0.03)**	**71%**	**72%**	**71%**
KNN	**82% (+/- 0.05)**	**80%**	**80%**	**80%**
Logistic Regression	**80% (+/- 0.03)**	**72%**	**69%**	**70%**
Random Forest	**78% (+/- 0.04)**	**77%**	**75%**	**76%**
Gaussian Naïve Bayes	**76% (+/- 0.05)**	**74%**	**74%**	**74%**
Gradient Boosting Trees	**80% (+/- 0.03)**	**77%**	**77%**	**77%**

Table 4.1.2: Results of approach-1 implementation

APPROACH 2 – WITH MFCC COEFFICIENTS

Among all the features retrieved from the audio signal, several researchers suggest that the MFCC values alone closely relate to the emotional tone of the audio.

Studies suggest that using the MFCC features can reduce the dimensionality of the training set, and thereby take less computation time.

This experiment repeats the same procedure as the previous but using only the MFCC values which are 13 dimensional.

The results of the second approach had scores lower than the first approach. The average accuracy of the models built using the second approach is 72%. KNN is the winning algorithm for this approach with a leading score of 80%+.

ALGORITHM	ACCURACY	PRECISION	RECALL	F1 SCORE
SVM	**79% (+/- 0.03)**	**78%**	**77%**	**77%**
Decision Tree	**68% (+/- 0.03)**	**65%**	**65%**	**65%**
KNN	**84% (+/- 0.05)**	**80%**	**80%**	**80%**
Logistic Regression	**80% (+/- 0.03)**	**78%**	**77%**	**77%**
Random Forest	**71% (+/- 0.04)**	**70%**	**69%**	**70%**
Gaussian Naïve Bayes	**75% (+/- 0.05)**	**73%**	**73%**	**72%**
Gradient Boosting Trees	**73% (+/- 0.03)**	**72%**	**69%**	**70%**

Table 4.1.3: Results of approach-2 implementation

APPROACH 3 – USING PCA FOR DECOMPOSITION

From the previous experiments, it can be learned that reducing the dimensions by cutting off the features, will reduce the performance of the model. This is because by cutting off features most of the information from the original dataset is not retained.

Principal Component Analysis (PCA) is a dimensionality reduction methodology in which the variation from the raw dataset can be specified to be retained.

PCA has a pre-requisite that the data should be standardized before performing PCA on it. For this experiment, a scree plot was plotted to determine the optimal number of components to be retained.

The steps of model building and evaluation was carried out next

1. The data is split into training and validation sets.
2. All the numeric values of the data are standardized to maintain a uniform distribution
3. PCA is performed on the data with 95% information retain value, and the respective number of components are observed.
4. The data is transformed to contain the principal components.
5. The model is built and evaluated with new data.

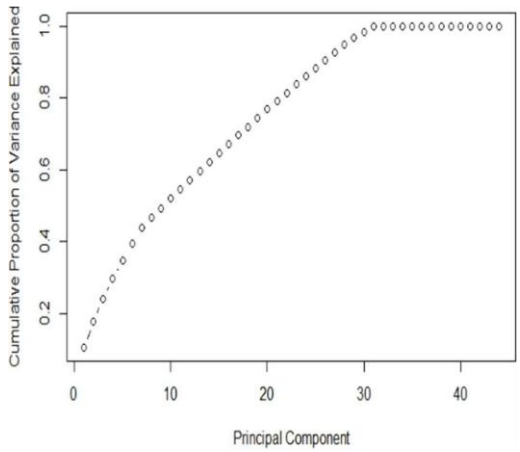

Fig 4.1.3 : Scree Plot

COMPARISION OF THE RESULTS

The various algorithms performed differently with each approach of the implementation. Since accuracy is not always a good measure for evaluating the model, the F1 scores can be used for comparison.

On comparing the F1 scores from the results of each approach, led to useful conclusions. SVM, KNN, and Logistic Regression classifiers have a low performance in the second approach and an improved performance in the third approach, as compared to the first approach.

The tree-based classifiers, such as Decision Tree, Random Forest, and Gradient Boosted trees performed the best for the first approach and has a low performance with the other approaches.

The KNN classifier had a constant performance score for all the three approaches.

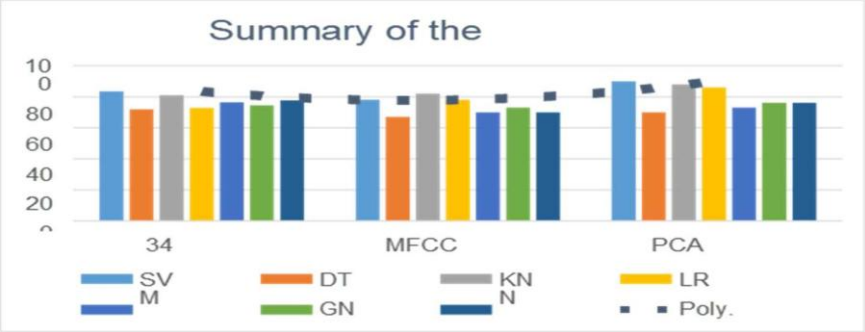

Fig 4.1.4 Summary Of experiments

4.2 Results

Several observations and conclusions can be derived from the results of the implementation. There is an overall improvement in the performance scores between the different approaches. The implementation following the first approach has a fair performance with a high score of 83% using the SVM algorithm, and the second approach worked well using the KNN algorithm for a high score of 80%, and the third approach had a 90% score using the SVM algorithm, which is the highest among all three approaches. The following observations can be made from the results:

Observation 1:

The improved scores of the SVM and KNN algorithm can be attributed to the dimensionality reduction used in the third approach. Reducing the dimensionality of the data increases the ratio of the size of the dataset to the number of dimensions, which reduces the bias of the classifier towards any particular class.

Contrastingly, the performance of the tree-based algorithms improves with a larger feature set.

This is because, the depth of the decision tree increases with adding more features, and thereby help making more accurate decisions.

The Bayesian principle of the Naïve Bayes algorithm works on the prior probability value calculated for each data in the training set, which remains unchanged among the two approaches, and hence the scores remain unchanged.

Observation 2 :

The overall performance of the second approach was lower than the other two approaches.

This is because of the selective feature approach fails to contain most of the information from the speech signal, and can be concluded that using only the MFCC values alone cannot be a good measure to classify the emotional content of speech.

Observation 3 : The classification report of the first approach shows that the misclassification is higher for the emotions Happy and Surprise.

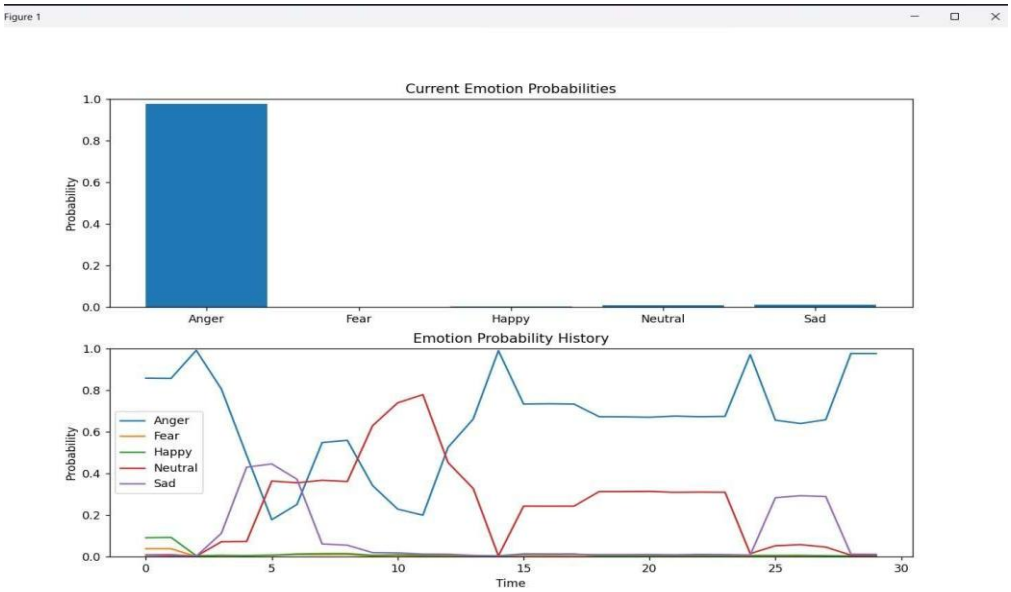

Fig 4.2.1: Evaluation of emotions via voice inputs

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

The emerging growth and development in the field of AI and machine learning have led to the new era of automation.

Most of these automated devices work based on voice commands from the user. Many advantages can be built over the existing systems if besides recognizing the words, the machines could comprehend the emotion of the speaker (user).

Some applications of a speech emotion detection system are computer-based tutorial applications, automated call center conversations, a diagnostic tool used for therapy and automatic translation system.

In this thesis, the steps of building a speech emotion detection system were discussed in detail and some experiments were carried out to understand the impact of each step.

Next, several novel approaches to feature extraction had been proposed in the earlier works, and selecting the best approach included performing many experiments.

Finally, the classifier selection involved learning about the strength and weakness of each classifying algorithm with respect to emotion recognition. At the end of the experimentation, it can be concluded that an integrated feature space will produce a better recognition rate when compared to a single feature.

For future advancements, the proposed project can be further modeled in terms of efficiency, accuracy, and usability.

Additional to emotions, the model can be extended to recognize feelings such as depression and mood changes. Such systems can be used by therapists to monitor the mood swings of the patients.

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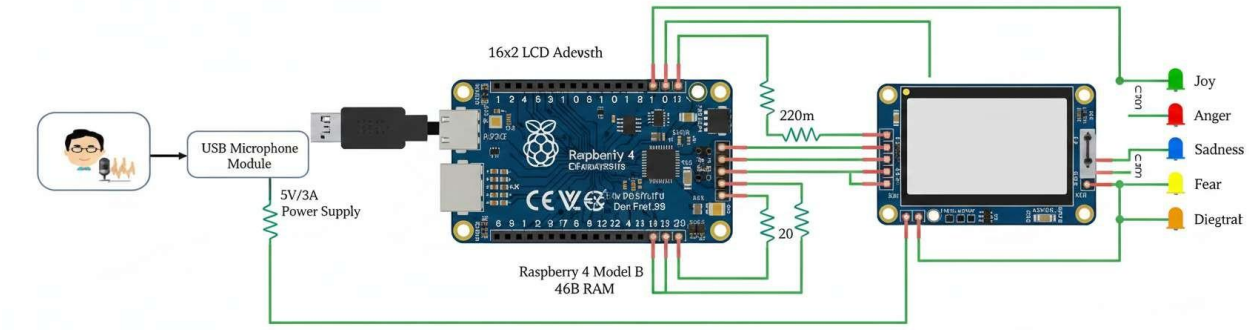
APPENDIX

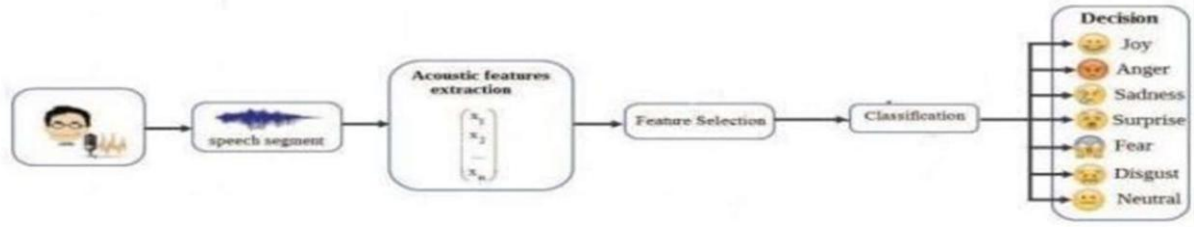
APPENDIX A

Circuit Diagram , Schematic and Components List

I. Circuit Diagram



II. Circuit Schematic Layout



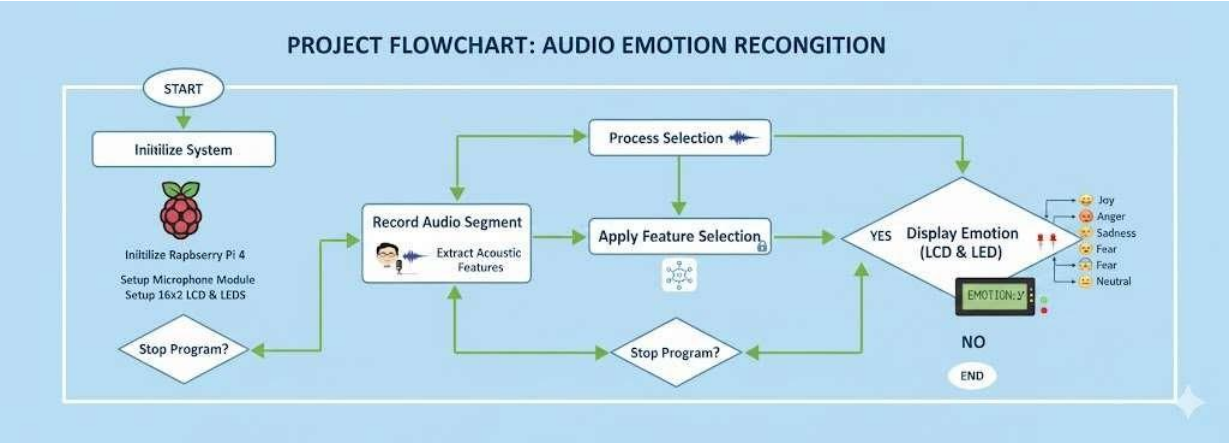
III. Components List :

1. Raspberry Pi 4 4GB RAM
2. Raspberry Pi Plug and Play Microphone Module
3. 16 X 2 LCD Display
4. LED Bulbs
5. Jumper wires

APPENDIX B

Flow chart and Code

I. Flow Chart :



II. Code :

Real-time speech emotion recognition working code :

```
import sounddevice as sd
import numpy as np
import librosa
from tensorflow.keras.models import load_model
import matplotlib.pyplot as plt
from collections import deque
import time
import threading
import queue

# Audio parameters
DURATION = 3  # seconds
SAMPLE_RATE = 16000
N_MFCC = 40
MAX_LEN = 200

# Model parameters
MODEL_PATH = "emotion_model_fixed_v3.keras"
EMOTIONS = np.load("label_encoder_classes.npy")
SMOOTHING = 5  # Increased smoothing window
CONFIDENCE_THRESHOLD = 0.15  # 15% threshold for showing secondary emotions

# Audio buffer for continuous recording
audio_buffer = queue.Queue()
recording = True

def audio_callback(indata, frames, time, status):
```

```
    if status:
        print(f"Status: {status}")
    audio_buffer.put(indata.copy())

def extract_features(y):
    """Extract MFCC features matching the training preprocessing."""
    try:
        # Normalize audio
        y = y / (np.max(np.abs(y)) + 1e-9)

        # Extract MFCCs and deltas
        mfcc = librosa.feature.mfcc(y=y, sr=SAMPLE_RATE, n_mfcc=N_MFCC)
        delta = librosa.feature.delta(mfcc)
        delta2 = librosa.feature.delta(mfcc, order=2)
        feat = np.vstack([mfcc, delta, delta2]).T

        # Pad or trim to MAX_LEN
        if feat.shape[0] < MAX_LEN:
            feat = np.pad(feat, ((0, MAX_LEN - feat.shape[0]), (0, 0)), mode='constant')
        else:
            feat = feat[:MAX_LEN, :]

        # Normalize features
        feat = (feat - np.mean(feat, axis=0)) / (np.std(feat, axis=0) + 1e-9)
        return feat[np.newaxis, ...]
    except Exception as e:
        print(f"Error in feature extraction: {str(e)}")
        return None

# Load model
print("Loading model...")
```

```
model = load_model(MODEL_PATH)

# Setup visualization
plt.ion()
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 8))

# Bar plot for current prediction
bars = ax1.bar(EMOTIONS, [0]*len(EMOTIONS))
ax1.set_ylim(0, 1)
ax1.set_ylabel("Probability")
ax1.set_title("Current Emotion Probabilities")

# Line plot for emotion history
history_length = 50
emotion_history = {emotion: deque([0]*history_length, maxlen=history_length) for
emotion in EMOTIONS}
lines = {}
for emotion in EMOTIONS:
    line, = ax2.plot(range(history_length), emotion_history[emotion], label=emotion)
    lines[emotion] = line
ax2.set_ylim(0, 1)
ax2.set_xlabel("Time")
ax2.set_ylabel("Probability")
ax2.legend()
ax2.set_title("Emotion Probability History")
plt.tight_layout()

# Initialize smoothing queue
recent_preds = deque(maxlen=SMOOTHING)

def update_plots(probs):
```

```
    # Update bar plot
    for bar, p in zip(bars, probs):
        bar.set_height(p)

    # Update line plot
    for i, emotion in enumerate(EMOTIONS):
        emotion_history[emotion].append(probs[i])
        lines[emotion].set_ydata(emotion_history[emotion])

    fig.canvas.draw()
    fig.canvas.flush_events()

print("\nStarting real-time emotion recognition...")
print("Available emotions:", EMOTIONS)
print("Press Ctrl+C to stop.\n")

try:
    with sd.InputStream(channels=1,
                        samplerate=SAMPLE_RATE,
                        callback=audio_callback):
        while recording:
            # Get audio from buffer
            audio_chunks = []
            start_time = time.time()

            while time.time() - start_time < DURATION:
                if not audio_buffer.empty():
                    chunk = audio_buffer.get()
                    audio_chunks.append(chunk)

            if not audio_chunks:
```

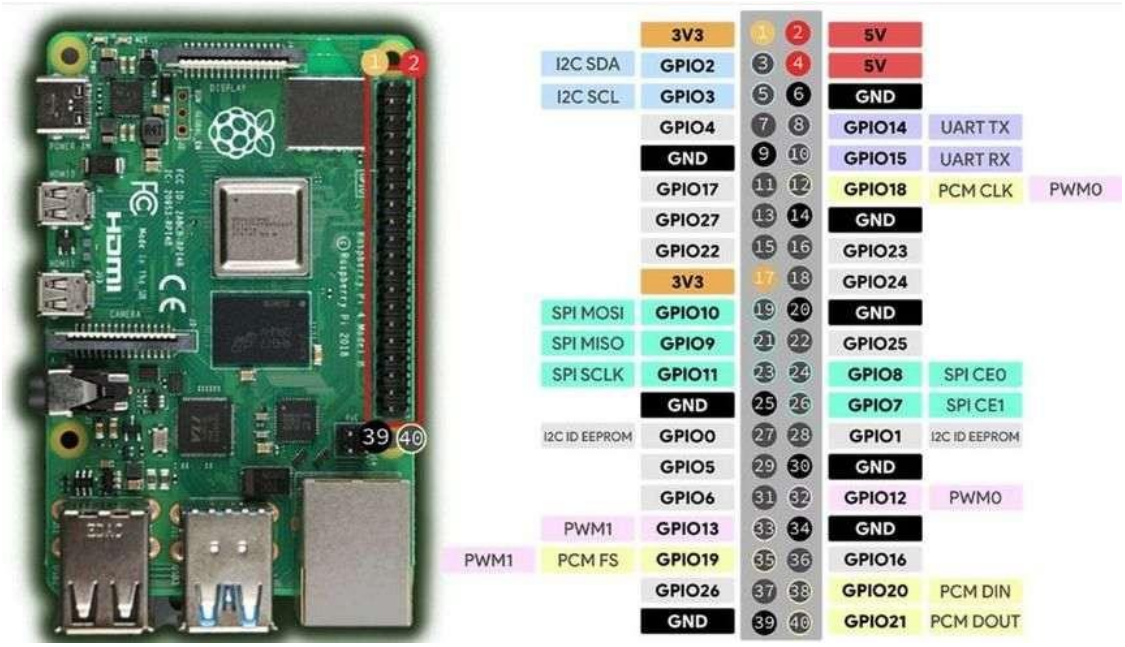
```
            continue

        # Combine chunks and flatten
        audio = np.concatenate(audio_chunks).flatten()

        # Skip if audio is too quiet
        if np.max(np.abs(audio)) < 0.01:
            print("No audio detected, skipping...")
            continue

        # Extract features
        features = extract_features(audio)
        if features is None:
            print("Failed to extract features, skipping...")
            continue

        # Predict emotion
        pred = model.predict(features, verbose=0)[0]
        recent_preds.append(pred)

        # Average predictions for smoothing
        smoothed = np.mean(recent_preds, axis=0)
        emotion_idx = np.argmax(smoothed)

        # Calculate confidence scores
        confidence = smoothed[emotion_idx] * 100

        # Print results
            print(f"\nPredicted emotion: {EMOTIONS[emotion_idx]} ({confidence:.1f}%
confidence)")
```

```
        # Show all emotions above threshold
        for idx, prob in enumerate(smoothed):
            if prob > CONFIDENCE_THRESHOLD:
                print(f"{EMOTIONS[idx]}:  {prob*100:.1f}%")

        # Update visualization
        update_plots(smoothed)

except KeyboardInterrupt:
    recording = False
    print("\nStopped.")
    plt.close()
```

APPENDIX C

Data Sheets

1. **Raspberry Pi 4 4GB RAM :**
   - The diagram of the 40-pin GPIO header is essential, showing the pin numbers, their functions (power, ground, GPIO, I2C, SPI, etc.), and voltage levels.
   - It takes the raw audio input from the microphone, processes it (e.g., performs Fast Fourier Transforms or speech recognition), and makes decisions based on the analysis.

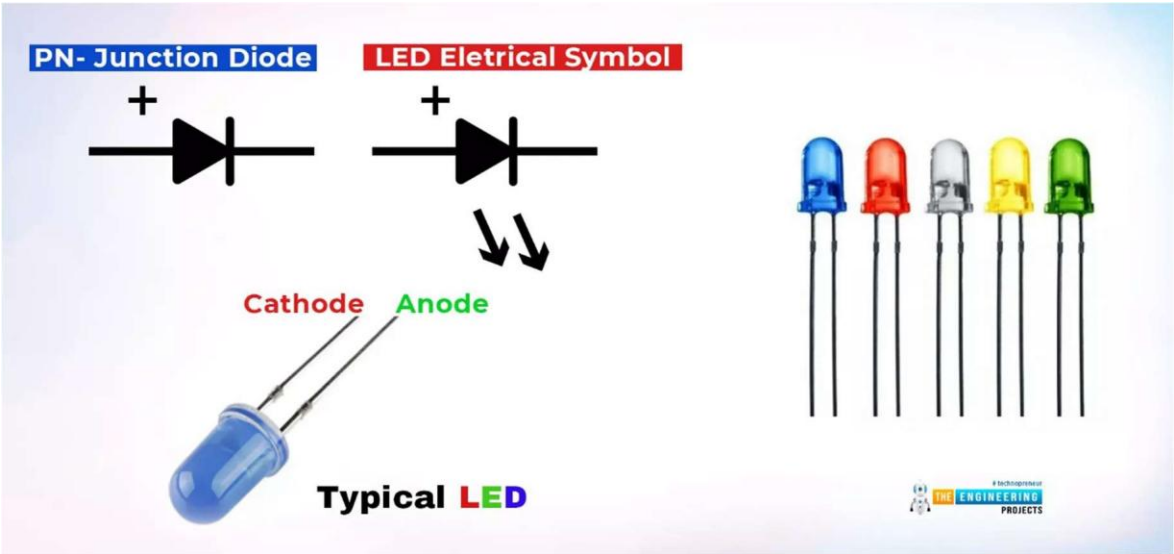


2. Raspberry Pi Plug and Play Microphone Module :

- Depending on the module, it either provides an analog signal (which needs an Analog-to-Digital Converter, or ADC) or a digital signal (like I2S or USB) that the **Raspberry Pi** can directly read and interpret as audio data.

- It converts sound pressure waves (vibrations in the air) into a tiny **electrical signal**.

3. LED



4. 16 X 2 LCD Display :
   - It provides a simple, low-power display to show messages or numerical data.
   - Showing the **status** of the system
     (e.g., "Listening...", "Command Recognized: ON").

APPENDIX D

IEEE Base Papers

2024 IEEE International Conference on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS)
Karnataka, India. Jun 28-29, 2024

2024 IEEE International Conference on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS) | 979-8-3503-8269-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICITEICS61368.2024.10625503

Deep Learning based Speech Emotion Recognition using Multiple Acoustic Features

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***Abstract*—Automatic speech emotion recognition (ASER) is crucial for multiple robotics applications, social media platforms, human-machine interaction systems, auto-chatbots, etc. However, the ASER performance is poor due to inadequate representation of the emotional speech, poor feature distinctiveness, and less generalization capability. This paper presents ASER based on a Deep Convolutional Neural Network (DCNN) that accepts the multiple acoustic features (MAF) to describe emotional speech. Further, it uses a combination of hybrid particle swarm optimization and Archimedes optimization algorithm (PSO-AOA) for selecting the essential features from MAFs. The outcomes of the MAF-PSO-AOA-DCNN are assessed on the BAUM dataset. It provides an overall accuracy of 95.20%, precision of 0.96, recall of 0.94, and F1-score of 0.95, which is superior to traditional techniques.**

***Keywords*—Speech Emotion Recognition, Multiple Acoustic Features, Deep Convolution Neural Network, Particle Swarm Optimization, Archimedes Optimization Algorithm**

I. Introduction

Automatic Speech emotion recognition (ASER) is a larger field concerned with automatically recognizing and categorizing emotional states represented via speech signals. It is a subfield of affective computing. Recently, SER has garnered much interest because it has the potential to be used in a wide variety of domains, including customer service, monitoring mental health, human-computer interaction, and social media sentiment analysis, amongst others [1-3].

The process of recognizing valuable information from speech signals in their most fundamental form is called voice recognition (SER). Qualities such as rhythm, pitch, intensity, and spectral characteristics are included in this category. These characteristics are categorized into different emotional categories using machine learning approaches. Various fundamental emotions may be found in these categories, including happiness, sadness, anger, fear, and neutrality. More complicated models, on the other hand, could account for more subtle emotional elements [4-7].

The SER procedure is broken down into a number of steps as it moves along. In the first place, voice samples are subjected to preprocessing to eliminate noise and emphasize relevant aspects. Following that, the acoustic characteristics of speech are captured via feature extraction techniques, and these recordings are then employed as input for machine learning models. SER uses several well-known machine learning methods, including support vector machines, recurrent neural networks, deep neural networks, and Gaussian mixture models, among others [8-10].

When training these algorithms, it is necessary to include labeled speech samples with the appropriate emotional characteristics in the datasets used for training. These datasets are intended to be collected to offer the model direction on recognizing patterns that indicate certain emotional states. On the other hand, the subjectivity and variety of emotional expression across various persons and cultural circumstances is a significant obstacle that must be overcome in social-emotional learning (SER). During the training process, it is essential to use datasets that are both varied and representative of the population [11-12].

Despite these challenges, SER has made substantial progress in recent years, resulting in more accurate models that can differentiate between a wide range of emotional states. For example, the investigation of consumer sentiment in contact centers, the monitoring of patients' emotional well-being in medical stings, and the development of user engagement via virtual assistants are all examples of applications that may be found in a wide variety of sectors. The subject of ASER is fast evolving and has significant promise for improving human-computer interaction and finding hidden emotional indicators in speech signals. Moreover, the field is seeing rapid development. It is necessary to engage in continual research and development operations to fully exploit the potential of SER across a wide variety of applications and industrial sectors [13].

This paper presents ASER based on the multiple acoustic features. The main contributions of the paper are summarized as follows:

- Implementation of the MAF consisting of spectral temporal and voice quality features.
- Feature selection using particle swarm optimization algorithm to select the crucial features
- Exploration and exploitation balancing in PSO using Archimedes optimization algorithm
- The performance is evaluated on the speech samples from the BAUM dataset based on accuracy, recall, precision, and F1-score.

The rest of the article is arranged as follows: Section II describes the detailed methodology. Section III provides the experimental results and augmentation discussions on the results. Further, section IV depicts the conclusion and gives the direction for future improvements.

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II. Related Work

Various DL-based schemes have been explored for the ASER. The ASER schemes based on the DL algorithm accept the raw speech or handcrafted features in 1D and 2D format. The 1D representation may include the traditional features derived from some statistical or single processing techniques. It concisely depicts emotional speech, but the performance relies on the features' type, size, and uniqueness. The 2D representation is more famous for speech emotion depiction because of its ability to provide information regarding frequency and time domain variations over the local part of the speech. However, 1D features needs 1D DL framework and 2D features requires 2D DL frameworks which are widely and easily available because of vast growth of DL for image processing applications [14-17].

The convolution-based DL frameworks describe the spatial and spectral characteristics of emotional speech. It provides the correlation and connectivity in global and local attributes of the speech. The DCNN has been implemented for one and multiple-dimensional data and shows huge generalization capability compared with other techniques. Many researchers have used long short-term memory to depict long-term representation of the speech signal and help to provide better time domain representation. Researchers in [2] developed the ASER based on MAFs and simple DCNN. However, the outcomes of the systems are limited because of redundancy in features and consideration of less salient features. Their approach used distinct frequency-domain, time-domain, and voice quality-based features to describe the unique characteristics of the emotion sound [18-19]

III. Methodology

The flow diagram of the ASER methodology is shown in Fig. 1. It encompasses speech signal acquisition, signal preprocessing, feature extraction, feature selection, and emotion recognition. The speech signals are taken from the BAUM dataset. The average filtering minimizes the speech's irregularities, noise, and artifacts. It standardizes the speech signal by considering the local window over the signal.

Fig. 1. Flow diagram of the proposed ASER

A. Multiple Acoustic Features

The MAFs include various spectral (SD), time domain (TD), and voice quality (VQ) features for the depiction of the impact of emotion on speech. The spectral domain features represent the spectral properties of the emotional speech.

The MFCC provides the energy and variation over the local pitch of the signal. It depicts the emotional impact on the regional characteristics of the speech. LPCC provides emotion-specific attributes of the speech. The spectral kurtosis and skewness (SK) provide the spectral domain changes in speech spectrogram obtained using scale invariant

TABLE I. Details about MAFs

Type of Features	Speech Features	Number of Features
Spectral Features	MFCC	13
	MFCC-Δ	13
	MFCC-ΔΔ	13
	LPCC	12
	Formant	01
	Mean of Formant	01
	Spectral kurtosis	257
	Skewness	01
	WPT features	56
Time domain features	Pitch frequency	01
	ZCR	01
	Hjorth Parameters	03
Voice Quality Features	Jitter	01
	Shimmer	01
Total Speech Features		374

Fourier transform. The speech signal is decomposed to the 3rd level using a wavelet packet transform to split the speech into frequency components that generate 8 packets. For every packet, mean, median, energy, kurtosis, skewness, standard deviation, and variance are calculated to characterize the effect of emotion on speech. The ZCR provides the noiseness in speech due to emotions. The pitch frequency (PF) provides maximum speech frequency where maximum speech content is accumulated. The Hjorth's parameters provide the speech signal's mobility, complexity, and activity. The jitter and shimmer provide the disparity in the time and amplitude domain. Jitter and shimmer depict the loudness of the speech [2].

$$Feat_{SD} = \{MFCC, LPCC, Formant, SK, WPT\} \quad (1)$$

$$Feat_{TD} = \{PF, HP, ZCR\} \quad (2)$$

$$Feat_{VQ} = \{Jitter, \quad Shimmer\} \quad (3)$$

$$Feat_{MAF} = \{Feat_{SD}, \quad Feat_{TD}, Feat_{VQ}\} \quad (4)$$

B. PSO-AOA-based Feature Selection

The PSO-AOA-based feature selection scheme's flow chart is represented in Fig. It uses the PSO algorithm for the feature selection, where the swarm population is considered analogous to the features. The PSO algorithm considers the possible combination of features as the initial swarm population. It computes the fitness for the every swarm based on mean, entropy and ratio of interclass to intra-class variability features (RI). The fitness is calculated using equation 4. The values of w1, w2, and w3 are selected such that w1+w2+w3=1.

$$Fit = w_1 * Entropy + w_2 * mean + w_3 * RI \quad (5)$$

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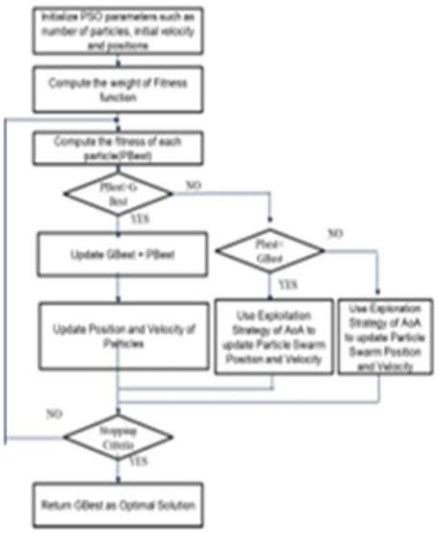

Fig. 2. Feature selection using PSO-AOA

The AOA algorithm is used to update the swarm population considering the collision of object strategy and no collision of object strategy. AOA is based on the Archimedes optimization algorithm, which is based on the principle that when an object is inserted into a fluid, it displaces it. The amount of the displaced fluid is equal to the Buoyance force applied to the object. If the rule is not satisfied, it updates the object's density, acceleration, and mass. These parameters are updated using two strategies, which consider collision of the objects and no collision of objects. Collision refers to exploitation, and no-collision relates to exploration. The exploitation considers the best objects obtained during the previous iteration for updating current object parameters. However, the exploration considers the random parameters for updating the current object positions.

A. DCNN for Feature Representation and Emotion Classification

The detailed flow diagram of proposed DCNN is illustrated in Fig.

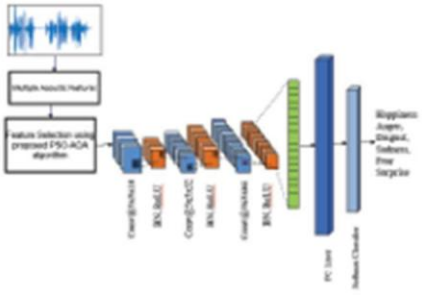

Fig. 3. DCNN for Feature Representation of SER

The DCNN consists of three layers for the representation of hierarchical features. It consists of convolution, rectified linear unit layer, and maximum pooling later for the deep feature representation. The convolution layer provides the correlation and connectivity between the different global and local features of the MAFs. The ReLU layer increases the features' nonlinear characteristics and improves accuracy. The equations for the convolution and ReLU are given in equation 6-8.

$$q(n) = MAF(n) \times CF(n) = \sum_{m=0}^{i-1} MAF(m).CF(n-m) \quad (6)$$

$$q_i^l = \sigma\left(b_i^l + \sum_j q_j^{l-1} \times CF_{ij}^l\right) \quad (7)$$

$$\sigma(z) = \max(0, q) \quad (8)$$

Here, CF stands for convolution filter, q provides convolution output, σ signifies ReLU output and b denotes bias value.The maximum pooling layer selects the salient features from the ReLU layer output and helps to minimize training time. The fully connected layer connects each neuron to increase its representation power. Finally, the softmax classifier is used for emotion recognition.

IV. Experimental Results and Discussions

The proposed system uses MATLAB on a computer with 20 GB RAM and a Windows operating environment. The DCNN is trained using the ADAM optimization algorithm for the 200 epochs and 600 iterations, with an initial learning rate of 0.001 and a batch size of 32. The accuracy, precision, recall, and F1 score are given by equation 9-12.

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

$$Recall = \frac{TN}{TN + FN} \quad (10)$$

$$Accuracy(\%) = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \quad (11)$$

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (12)$$

The experimental results for the suggested ASER scheme are illustrated in Table II-IV.

TABLE II. Accuracy for Different Algorithm

Method	Accuracy	Recall	Precision	F1-Score
MAF+DCNN	93.20	0.90	0.93	0.91
MAF+PSO+DCNN	93.60	0.91	0.4	0.92
MAF+PSO+AOA+DCNN	95.20	0.94	0.96	0.95

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TABLE III. ACCURACY COMPARISON FOR ML AND DL SCHEME

Method	Accuracy
MAF+PSO+AOA+SVM	93.60
MAF+PSO+AOA+KNN	93.80
MAF+PSO+AOA+DCNN	95.20

The algorithm's performance is compared with the ASER with and without feature selection. It is noted that the feature selection helps to enhance the distinguishing capability of the MAFs. It reduces the dimensions of the features. The feature selection assists in minimizing the redundant, non-important features from the available features. This decreases the computational complexity of the DCNN model and achieves better accuracy, as given in Table II. When the DCNN is trained for the MAFs only, it delivers an accuracy of 93.20%, recall of 0.90, precision of 0.93, and F1-score of 0.91. The MAF-PSO-DCNN offers delivers an accuracy of 93.60%, recall of 0.91, precision of 0.94, and F1-score of 0.92. The PSO-based feature selection provides better results than MAFs. Further, the MAF-PSO-DCNN offers an accuracy of 95.20%, recall of 0.94, precision of 0.96, and F1-score of 0.95. The PSO-based feature selection provides better results than MAFs.

TABLE IV. ACCURACY FOR DIFFERENT FEATURES

No of features	Accuracy
50	86.20
100	90.00
150	93.20
200	94.60
250	95.20
300	95.10
350	94.80

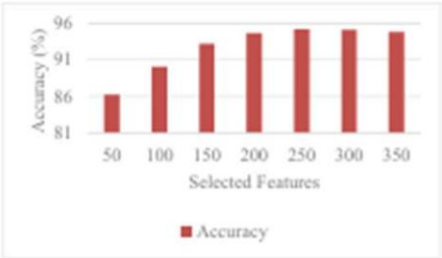

Fig. 4. Accuracy for various features

The outcomes of the ASER systems are also evaluated for the traditional Machine learning classifiers such as K-Nearest Neighbor (KNN) and support vector machine (SVM). It offers 93.60% accuracy for SVM and 93.80% for the KNN classifier. However, the DCNN helps to improve the hierarchical representation of the MAFs and provides improved accuracy of 95.20% for 250 features. Table III provides better results for the 250 features. Increasing features up to 250 shows a boost in accuracy. However, after 250, the results show stagnation. Therefore, 250 selected as input to DCNN algorithm for training.

V. CONCLUSIONS AND FUTURE SCOPES

This paper provides the implementation of DCNN-based ASER. The MAFs provide the impact of the human sentiment on the frequency, time, and voice quality characteristics. The DCNN enhances the distinguishing capability of the different global and local features. The PSO-AOA provides imperative features based on the feature's high variance, entropy, and distinctiveness. The AOA algorithm balances the exploration and exploitation phenomenon in conventional PSO by updating the population to consider collision and non-collision of objects. The MAF-PSO-DCNN offers an accuracy of 95.20%, recall of 0.94, precision of 0.96, and F1-score of 0.95. It aids in declining the computational intricacy of the ASER system. The proposed model shows better generalization capability. It tackles the effect of variations in speech signals due to language, intonation, gender, religious, and regional influences on the language. In the future, the performance of the suggested ASER system can be boosted by optimizing the PSO-AOA feature selection algorithm.

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HKBK COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAM OBJECTIVES (POs)

PO – 1 : Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO – 2 : Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO – 3 : Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO – 4 : Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO – 5 : Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO – 6 : The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO – 7 : Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of need for sustainable development.

PO – 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO – 9 : Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO – 10 : Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO – 11 : Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO – 12 : Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

HKBK COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

VISION AND MISSION OF THE INSTITUTE

VISION

To empower students through wholesome education and enable the students to develop into highly qualified and trained professionals with ethics and emerge as responsible citizens with broad outlook to build a vibrant nation.

MISSION

- To achieve academic excellence through in-depth knowledge in science, engineering and technology through dedication to duty, innovation in teaching and faith in human values.
- To enable our students to develop into outstanding professionals with high ethical standards to face the challenges of the 21" century.
- To provide educational opportunities to the deprived and weaker section of the society, to uplift their socio-economic status.

VISION AND MISSION OF THE DEPARTMENT

VISION

To shape the students as disciplined human engineers who can build a strong, peaceful and vibrant country and focus on mutual respect, tolerance and professional ethics.

MISSION

M1: To provide the best possible educational experience.

M2: Excellence in teaching and research activities for today's students and professionals of tomorrow.

M3: To hone young minds and train them to be conscientious individuals.

M4: Will serve the society as competent professionals in the field of Electronics and Communication Engineering.

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